

Question 1

1. An investor is interested in allocating his wealth across the following two securities: Apple (AAPL) and Microsoft (MSFT).
 - (a) Using the R output in Figure 1(a) on AAPL (R_AAPL) and MSFT (R_MSFT) returns respectively, what percentage of the investor's wealth would you advise him to allocate to each security?
 - (b) Verify your answer to Q1(a) by making use of the R output in Figure 1(b) and the optimal weights formulae derived from the risk minimisation problem applied to the portfolio consisting of AAPL and MSFT.
2. Consider the following Fama-French four factor model

$$r_t - r_{ft} = \beta_0 + \beta_1(r_{mt} - r_{ft}) + \beta_2\text{SMB}_t + \beta_3\text{HML}_t + \beta_4\text{MOM}_t + \eta_t, \quad (1)$$

where r_t denotes monthly returns on the Fama-French durables (DUR) portfolio (includes consumer durables – Cars, TVs, Furniture, Household Appliances), $(r_{mt} - r_{ft})$ is the excess market return (the market factor - MKT-RF), r_{ft} is the risk free interest rate, SMB is the size factor, HML is the value factor, and MOM is the momentum factor. OLS output from estimating (1) over the period January 2005 to February 2023 (a total of 218 observations) is shown in Figure 2(a).

- (a) Show the analytical decomposition of total risk into systematic and idiosyncratic risks derived from (1), if we assume that $\beta_0 = \beta_2 = \beta_3 = \beta_4 = 0$ and r_{ft} is time invariant.
 - (b) Using the R output in Figure 2(a), report what fraction of total risk implied by (1) is systematic and what fraction is idiosyncratic.
3. Using the R output of regression (1) in Figure 2(a):
 - (a) Interpret the coefficient estimate of the SMB factor. Test whether this coefficient is statistically significant at the 1% significance level. Briefly comment on your conclusion.
 - (b) Test whether the one factor CAPM model is preferred over (1) using the information provided in Figures 2(a) and 2(b). What conclusion do you draw?

(continued)

```

Call:
lm(formula = R_AAPL ~ I(R_AAPL - R_MSFT))

Residuals:
    Min       1Q   Median       3Q      Max
-0.36581 -0.04581  0.00509  0.04585  0.33269

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.01627    0.00409   3.977 8.31e-05 ***
I(R_AAPL - R_MSFT) 0.78408    0.03349  23.412 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0815 on 395 degrees of freedom
Multiple R-squared:  0.5812,    Adjusted R-squared:  0.5801
F-statistic: 548.1 on 1 and 395 DF,  p-value: < 2.2e-16

```

Figure 1(a)

```

> lr = merge(R_AAPL,R_MSFT)
> colnames(lr) <- c("R_AAPL", "R_MSFT")
> cov.mat = cov(lr)
> cov.mat
      R_AAPL      R_MSFT
R_AAPL 0.015820443 0.004093765
R_MSFT 0.004093765 0.007323076

```

Figure 1(b)

```

> fit.1 <- lm(DUR-RF~`Mkt-RF`+SMB+HML+MOM, data = data.xts)
> summary(fit.1)

Call:
lm(formula = DUR - RF ~ `Mkt-RF` + SMB + HML + MOM, data = data.xts)

Residuals:
    Min       1Q   Median       3Q      Max
-15.9094  -2.6695  -0.5062   1.8248  28.5578

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.06119    0.36710  -0.167 0.867768
`Mkt-RF`     1.46097    0.09099  16.056 < 2e-16 ***
SMB           0.54912    0.15918   3.450 0.000676 ***
HML          -0.42110    0.12338  -3.413 0.000769 ***
MOM          -0.40285    0.09159  -4.398 1.72e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.33 on 213 degrees of freedom
Multiple R-squared:  0.6951,    Adjusted R-squared:  0.6893
F-statistic: 121.4 on 4 and 213 DF,  p-value: < 2.2e-16

```

Figure 2(a)

```

> fit.2 <- lm(DUR-RF~`Mkt-RF`, data = data.xts)
> summary(fit.2)

Call:
lm(formula = DUR - RF ~ `Mkt-RF`, data = data.xts)

Residuals:
    Min       1Q   Median       3Q      Max
-19.5116  -2.6201  -0.4685   2.0639  28.2819

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.2043     0.3929   -0.52    0.604
`Mkt-RF`      1.6879     0.0856   19.72 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.728 on 216 degrees of freedom
Multiple R-squared:  0.6429,    Adjusted R-squared:  0.6412
F-statistic: 388.8 on 1 and 216 DF,  p-value: < 2.2e-16

```

Figure 2(b)

Statistical tables are provided after the end of each question.

————— END OF QUESTION 1 —————

Question 2

Consider the daily price of Apple (AAPL) from 2 January 2020 to 30 April 2025. Let p_t denote the log price of AAPL.

1. Assume that we model p_t as:

$$p_t = \mu + p_{t-1} + \varepsilon_t \quad (2)$$

where $p_0 = 0$ and ε_t is white noise with mean 0 and variance σ^2 . The Augmented Dickey-Fuller (ADF) test results applied to p_t are provided in Figure 3(a). Write the null and alternative hypotheses that are implied by this test. What conclusion do you draw about the p_t process?

2. We compute the time series of log returns on Apple, $r_t = p_t - p_{t-1}$. Let F_{t-1} denote the information set available at time $t - 1$. Suppose r_t is stationary and follows an AR(1) process:

$$r_t = 0.1 + 0.5r_{t-1} + u_t \quad (3)$$

where u_t is a martingale difference sequence (MDS) and $\mathbb{V}(u_t|F_{t-1}) = 4$. Provide numerical answers to the following questions where applicable.

- (a) What is the conditional mean of r_t ?
 - (b) Using your answer above, what is the unconditional mean of r_t ?
 - (c) What is $\mathbb{V}(r_t|F_{t-1})$?
 - (d) Using your answers above, what is $\mathbb{V}(r_t)$?
 - (e) Compare your answers to parts (c) and (d). Are they as you would expect? Briefly explain.
3. Using Figure 3(b), write down the fitted model for r_t . Table 1 presents the observed return r_t , the residuals $\hat{u}_t$ obtained from Figure 3(b), and the fitted conditional variance, also obtained from Figure 3(b). Assuming that r_t follows a normal distribution, compute the 95% prediction interval for the one-step-ahead forecast (1 May 2025) of r_t .
 4. Using Figure 3(b) and Table 1, calculate the 1-day-ahead 5% VaR on the \$1m investment in AAPL stock.

(continued)

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression trend

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-0.140709 -0.009098  0.000153  0.010450  0.111558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.453e-02  2.212e-02   2.465   0.0138 *
z.lag.1      -1.014e-02  4.221e-03  -2.402   0.0164 *
tt           5.559e-06  3.055e-06   1.820   0.0691 .
z.diff.lag   -1.667e-01  2.712e-02  -6.148 1.04e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01904 on 1322 degrees of freedom
Multiple R-squared:  0.03401,    Adjusted R-squared:  0.03182
F-statistic: 15.51 on 3 and 1322 DF,  p-value: 6.386e-10

value of test-statistic is: -2.4021 2.8126 3.1557

critical values for test statistics:
      1pct  5pct 10pct
tau3  -3.96 -3.41 -3.12
phi2   6.09  4.68  4.03
phi3   8.27  6.25  5.34
```

Figure 3(a)

```

> spec_garch <- ugarchspec(
+   variance.model = list(
+     model = "sGARCH",
+     garchorder = c(1, 1)
+   ),
+   mean.model = list(
+     armaorder = c(1, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_garch <- ugarchfit(
+   spec = spec_garch,
+   data = log_ret*100
+ )
>
> fit_garch@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.08991742  0.03985154   2.256310 2.405123e-02
ar1     -0.07880947  0.02938161  -2.682272 7.312390e-03
omega    0.16341315  0.06154980   2.654974 7.931444e-03
alpha1   0.10380974  0.02434972   4.263282 2.014458e-05
beta1    0.84884904  0.03832817  22.146872 0.000000e+00

```

Figure 3(b)

Table 1			
	r_t (%)	$\hat{u}_t$	Fitted conditional variance
29 April 2025	0.734	0.6189	5.2336
30 April 2025	0.309	0.2660	4.6638

Statistical tables are provided after the end of each question.

————— END OF QUESTION 2 —————

Question 3

Nicky is interested in investing in Volkswagen stock. In order to understand the evolution of this security over time, she models its daily log returns, denoted by $r_{V,t}$, over the period 4 January 2000 to 24 April 2023.

1. She uses a constant mean equation and conducts an ARCH(5) test on the residuals of this model, $\hat{v}_t$. The results are shown in Figure 4.
 - (a) Provide all steps in the testing procedure. What conclusion do you draw?
 - (b) Why do you think Nicky conducted the ARCH test on residuals, $\hat{v}_t$, rather than the White test? Briefly explain the difference between the two tests. [Hint: do not write all steps of the testing procedure].
2. Nicky applies a GARCH(1,1) model to $r_{V,t}$. Results are shown in Figure 5.
 - (a) Neatly write the theoretical model that is implied by the results of Figure 5 and verify that the parameter estimates comply with the conditions of this model. Why are these conditions needed?
 - (b) Derive the unconditional skewness of the errors v_t from the theoretical model in Q3.2(a). What conclusion do you draw?
3. Nicky further applies a GJR-GARCH model to the Volkswagen returns equation, $r_{V,t}$.
 - (a) Neatly write the theoretical model corresponding to GJR-GARCH and briefly explain how this model differs from the GARCH(1,1) theoretical model.
 - (b) Which parameters from the GJR-GARCH model determine the curvature of its news impact curve (NIC)? Provide the numerical values of the NIC curvature using the information in Figure 6.

(continued)

```

> FintTS::ArchTest(res, lags = 5)

      ARCH LM-test; Null hypothesis: no ARCH effects

data:  res
Chi-squared = 1135.6, df = 5, p-value < 2.2e-16

```

Figure 4

```

> library(rugarch)
> spec_garch <- ugarchspec(
+   variance.model = list(
+     model = "sGARCH",
+     garchorder = c(1, 1)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_garch <- ugarchfit(
+   spec = spec_garch,
+   data = r_v
+ )
>
> fit_garch@fit$matcoef

```

	Estimate	Std. Error	t value	Pr(> t)
mu	6.960937e-04	2.406820e-04	2.892171	3.825891e-03
omega	8.996380e-06	1.820632e-06	4.941351	7.758329e-07
alpha1	8.078663e-02	4.117994e-03	19.617958	0.000000e+00
beta1	9.040081e-01	5.199816e-03	173.853878	0.000000e+00

Figure 5


```

> spec_gjr <- ugarchspec(
+   variance.model = list(
+     model = "gjrGARCH",
+     garchOrder = c(1, 1)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_gjr <- ugarchfit(
+   spec = spec_gjr,
+   data = r_v
+ )
>
> fit_gjr@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      4.705136e-04 2.374653e-04   1.981399 4.754652e-02
omega   1.019193e-05 9.335206e-07  10.917739 0.000000e+00
alpha1  5.486562e-02 3.663183e-03  14.977580 0.000000e+00
beta1   9.000092e-01 5.571515e-03 161.537625 0.000000e+00
gamma1  5.483184e-02 9.789800e-03   5.600916 2.132224e-08

```

Figure 6

Statistical tables are provided after the end of each question.

————— **END OF QUESTION 3** —————

Question 4

An investor runs the following Fama-French four factor model over the period January 2009 to December 2019:

$$r_{M,t} - r_{ft} = \beta_0 + \beta_1(r_{mt} - r_{ft}) + \beta_2\text{SMB}_t + \beta_3\text{HML}_t + \beta_4\text{MOM}_t + u_t, \quad (4)$$

where $r_{M,t}$ denotes monthly returns on the Microsoft security, $(r_{mt} - r_{ft})$ is the excess market return (the market factor - MKT-RF), r_{ft} is the risk free interest rate, SMB is the size factor, HML is the value factor, and MOM is the momentum factor.

1. Set up a test on the residuals in (4), $\hat{u}_t$, to verify whether the Efficient Market Hypothesis holds.
2. Based on the results of Figure 7 what conclusion do you draw for the selected time period?
3. What are the three forms of EMH? How do they differ from each other? Briefly explain.

```
> lmtest::bgtest(fit, order = 2, type = "chisq")  
  
Breusch-Godfrey test for serial correlation of order up to 2  
  
data: fit  
LM test = 64.641, df = 2, p-value = 9.191e-15
```

Figure 7

Statistical tables are provided after the end of each question.

————— **END OF QUESTION 4** —————

Question 5

A researcher is interested in modelling the Nikkei index daily log returns over the period from 5 January 2000 to 8 May 2020.

1. She uses an MA(1) model and considers an ARCH(5) specification for the variance equation. The estimation results from this model are shown in Figure 8. Write the theoretical model implied by the regression output in Figure 8.
2. She then uses a constant mean equation and considers an ARCH(5) specification for the variance equation. The estimation results from this ARCH(5) model are shown in Figure 9. Write the theoretical model implied by the regression output in Figure 9.
3. She subsequently considers a GARCH specification and re-estimates her model. The results are shown in Figure 10. Using Figures 9 and 10, evaluate which of the ARCH and GARCH models you would opt for. Explain your reasoning and the statistics that you used.

```
> spec_ma1arch5 <- ugarchspec(
+   variance.model = list(
+     model = "sGARCH",
+     garchOrder = c(5, 0)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 1),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_ma1arch5 <- ugarchfit(
+   spec = spec_ma1arch5,
+   data = lr*100
+ )
>
> fit_ma1arch5@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.05312799	0.01633806	3.251792	1.146797e-03
ma1	-0.02237429	0.01498448	-1.493164	1.353942e-01
omega	0.72948343	0.04097485	17.803199	0.000000e+00
alpha1	0.10741200	0.01677491	6.403132	1.522211e-10
alpha2	0.19922098	0.02155758	9.241342	0.000000e+00
alpha3	0.16710864	0.02090032	7.995508	1.332268e-15
alpha4	0.11113704	0.01763322	6.302708	2.924903e-10
alpha5	0.11595733	0.01828853	6.340440	2.291101e-10

Figure 8

```

> spec_arch5 <- ugarchspec(
+   variance.model = list(
+     model = "SGARCH",
+     garchorder = c(5, 0)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_arch5 <- ugarchfit(
+   spec = spec_arch5,
+   data = 1r*100
+ )
>
> fit_arch5@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.05351403  0.01669290   3.205797 1.346891e-03
omega   0.73097205  0.04094154  17.854042 0.000000e+00
alpha1  0.10732388  0.01680492   6.386456 1.697746e-10
alpha2  0.19810827  0.02146971   9.227337 0.000000e+00
alpha3  0.16846296  0.02095228   8.040318 8.881784e-16
alpha4  0.11093323  0.01764332   6.287547 3.225205e-10
alpha5  0.11503043  0.01835089   6.268385 3.648115e-10
>
> # information criteria AIC
> Ics.arch5 <- infocriteria(fit_arch5)
> Ics.arch5

Akaike      3.429337
Bayes       3.438487
Shibata     3.429333
Hannan-Quinn 3.432545

```

Figure 9

```

> spec_garch21 <- ugarchspec(
+   variance.model = list(
+     model = "SGARCH",
+     garchOrder = c(2, 1)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_garch21 <- ugarchfit(
+   spec = spec_garch21,
+   data = 1r*100
+ )
>
> fit_garch21@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.05305716  0.01642468  3.230331 1.236471e-03
omega   0.05127234  0.00960747  5.336716 9.464528e-08
alpha1  0.08213008  0.01427050  5.755235 8.652109e-09
alpha2  0.04339894  0.01846342  2.350536 1.874639e-02
beta1   0.85522831  0.01409044 60.695638 0.000000e+00
>
> ICs.garch21 <- infocriteria(fit_garch21)
> ICs.garch21

Akaike      3.386934
Bayes       3.393470
shibata     3.386932
Hannan-Quinn 3.389225

```

Figure 10

Statistical tables are provided after the end of each question.

————— END OF QUESTION 5 —————

STATISTICAL TABLES

TABLE G.1 Cumulative Areas under the Standard Normal Distribution

<i>z</i>	0	1	2	3	4	5	6	7	8	9
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776

(continued)

TABLE G.1 (Continued)

z	0	1	2	3	4	5	6	7	8	9
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Example: If $Z \sim \text{Normal}(0,1)$, then $P(Z \leq -1.32) = .0934$ and $P(Z \leq 1.84) = .9671$.

Source: This table was generated using the Stata® function `normprob`.

(continued)

TABLE G.2 Critical Values of the <i>t</i> Distribution						
		Significance Level				
1-Tailed:		.10	.05	.025	.01	.005
2-Tailed:		.20	.10	.05	.02	.01
D e g r e e s o f F r e e d o m	1	3.078	6.314	12.706	31.821	63.657
	2	1.886	2.920	4.303	6.965	9.925
	3	1.638	2.353	3.182	4.541	5.841
	4	1.533	2.132	2.776	3.747	4.604
	5	1.476	2.015	2.571	3.365	4.032
	6	1.440	1.943	2.447	3.143	3.707
	7	1.415	1.895	2.365	2.998	3.499
	8	1.397	1.860	2.306	2.896	3.355
	9	1.383	1.833	2.262	2.821	3.250
	10	1.372	1.812	2.228	2.764	3.169
	11	1.363	1.796	2.201	2.718	3.106
	12	1.356	1.782	2.179	2.681	3.055
	13	1.350	1.771	2.160	2.650	3.012
	14	1.345	1.761	2.145	2.624	2.977
	15	1.341	1.753	2.131	2.602	2.947
	16	1.337	1.746	2.120	2.583	2.921
	17	1.333	1.740	2.110	2.567	2.898
	18	1.330	1.734	2.101	2.552	2.878
	19	1.328	1.729	2.093	2.539	2.861
	20	1.325	1.725	2.086	2.528	2.845
	21	1.323	1.721	2.080	2.518	2.831
	22	1.321	1.717	2.074	2.508	2.819
	23	1.319	1.714	2.069	2.500	2.807
	24	1.318	1.711	2.064	2.492	2.797
	25	1.316	1.708	2.060	2.485	2.787
	26	1.315	1.706	2.056	2.479	2.779
	27	1.314	1.703	2.052	2.473	2.771
	28	1.313	1.701	2.048	2.467	2.763
	29	1.311	1.699	2.045	2.462	2.756
	30	1.310	1.697	2.042	2.457	2.750
	40	1.303	1.684	2.021	2.423	2.704
	60	1.296	1.671	2.000	2.390	2.660
	90	1.291	1.662	1.987	2.368	2.632
	120	1.289	1.658	1.980	2.358	2.617
	∞	1.282	1.645	1.960	2.326	2.576

Examples: The 1% critical value for a one-tailed test with 25 *df* is 2.485. The 5% critical value for a two-tailed test with large (> 120) *df* is 1.96.

Source: This table was generated using the Stata® function invttail.

(continued)

TABLE G.3a 10% Critical Values of the *F* Distribution

		Numerator Degrees of Freedom									
		1	2	3	4	5	6	7	8	9	10
D e n o m i n a t o r	10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32
	11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25
	12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19
	13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14
	14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10
	15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06
	16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03
	17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00
	18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98
	19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96
D e g r e e s	20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94
	21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92
	22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90
	23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89
	24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88
	25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87
	26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86
	27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85
	28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84
	29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83
F r e e d o m	30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82
	40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76
	60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71
	90	2.76	2.36	2.15	2.01	1.91	1.84	1.78	1.74	1.70	1.67
	120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65
	∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60

Example: The 10% critical value for numerator $df = 2$ and denominator $df = 40$ is 2.44.

Source: This table was generated using the Stata® function invFtail.

(continued)

TABLE G.3b 5% Critical Values of the <i>F</i> Distribution											
		Numerator Degrees of Freedom									
		1	2	3	4	5	6	7	8	9	10
D e n o m i n a t o r D e g r e e s o f F r e e d o m	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
F r e e d o m	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
	90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
	120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91
		∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.83

Example: The 5% critical value for numerator $df = 4$ and large denominator $df(\infty)$ is 2.37.

Source: This table was generated using the Stata® function invFtail.

(continued)

TABLE G.3c 1% Critical Values of the *F* Distribution

		Numerator Degrees of Freedom									
		1	2	3	4	5	6	7	8	9	10
D e n o m i n a t o r	10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85
	11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
	12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
	13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10
	14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94
	15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80
	16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69
	17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59
	18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51
	19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43
	20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37
	21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31
	22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26
	23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21
	24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17
	25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13
	26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09
	27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06
	28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03
	29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00
	30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98
D e g r e e s o f F r e e d o m	40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80
	60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
	90	6.93	4.85	4.01	3.54	3.23	3.01	2.84	2.72	2.61	2.52
	120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47
	∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32

Example: The 1% critical value for numerator $df = 3$ and denominator $df = 60$ is 4.13.

Source: This table was generated using the Stata® function invFtail.

(continued)

TABLE G.4 Critical Values of the Chi-Square Distribution

		Significance Level		
		.10	.05	.01
D e g r e e s o f F r e e d o m	1	2.71	3.84	6.63
	2	4.61	5.99	9.21
	3	6.25	7.81	11.34
	4	7.78	9.49	13.28
	5	9.24	11.07	15.09
	6	10.64	12.59	16.81
	7	12.02	14.07	18.48
	8	13.36	15.51	20.09
	9	14.68	16.92	21.67
	10	15.99	18.31	23.21
	11	17.28	19.68	24.72
	12	18.55	21.03	26.22
	13	19.81	22.36	27.69
	14	21.06	23.68	29.14
	15	22.31	25.00	30.58
	16	23.54	26.30	32.00
	17	24.77	27.59	33.41
	18	25.99	28.87	34.81
	19	27.20	30.14	36.19
	20	28.41	31.41	37.57
	21	29.62	32.67	38.93
	22	30.81	33.92	40.29
	23	32.01	35.17	41.64
	24	33.20	36.42	42.98
	25	34.38	37.65	44.31
	26	35.56	38.89	45.64
	27	36.74	40.11	46.96
	28	37.92	41.34	48.28
	29	39.09	42.56	49.59
	30	40.26	43.77	50.89

Example: The 5% critical value with $df = 8$ is 15.51.

Source: This table was generated using the Stata[®] function `invchi2tail`.